








# CIRP - Literature Review

## Customer Intelligence & Retention Platform

DEPI • AI & Machine Learning Track • Round 5

Graduation Project

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| **Project** | Customer Churn Prediction & Segmentation Platform (CIRP) |
| **Track** | DEPI - AI & Machine Learning |
| **Project Leader** | Mohamed Atef Mohamed Abdo |
| **Version / Date** | v1.0 - October 2026 |

# 1. Introduction

This review covers how churn is defined in retail, which methods are used for churn prediction and segmentation, and how they shape the design of CIRP.

# 2. Customer Churn in Retail

In retail the customer relationship is non-contractual, so churn needs an operational definition: no purchase within a fixed window after an observation date. CIRP labels a customer as churned if there is no order in the 90 days after the snapshot date.

The data is Online Retail II (a UK online retailer, 2009-2011). Many customers are wholesalers, so orders are large and irregular and most features are strongly skewed.

# 3. Machine Learning for Churn Prediction

## 3.1 Algorithms

Churn prediction is a binary classification problem. Comparative studies test the same classifier families and show that richer behavioral features improve results.

- **Logistic Regression:** fast and explainable through its coefficients; the standard baseline.
- **Decision Tree:** readable rules, but can overfit.
- **Support Vector Machine:** effective on scaled numeric features, but harder to explain.
- **Random Forest and Gradient Boosting:** tree ensembles that often give the best accuracy on tabular data.

CIRP compares Logistic Regression, Decision Tree and Random Forest; the tuned Random Forest was kept as the final churn model.

## 3.2 Evaluation and class imbalance

Accuracy hides poor results here: 61% of the 22,372 customer-snapshots are labelled churn, so the majority class alone already reaches 61% accuracy and accuracy does not show whether the model finds the right customers. The precision-recall curve is more informative than ROC under this imbalance, and Precision, Recall and F1 separate the two errors. Missing a churner costs more than an unnecessary offer, so CIRP reports Precision, Recall, F1, ROC-AUC and PR-AUC and chooses the threshold on validation data: for the Random Forest, lowering the threshold from 0.50 to 0.45 raised Recall from 0.914 to 0.943 and lowered Precision from 0.800 to 0.785, and 0.45 was kept.

## 3.3 Explainability, feature selection and leakage

Coefficients and tree feature importances show which factors drive churn; CIRP uses global feature importance.

CIRP selects features on training data only: correlation with the target, removal of pairs above 0.85 and an F-test ranking. This removed 3 of the 22 features (orders_last_30d, orders_last_60d, spend_trend)

and left 19, and the strongest single correlation with churn is 0.34, so the model relies on the combination of features rather than on one signal. Leakage gives over-optimistic results, so CIRP splits by time and uses the test period once.

# 4. Customer Segmentation

## 4.1 RFM features

RFM (Recency, Frequency, Monetary value) combined with K-Means is the classic approach to retail segmentation and produces actionable customer groups. CIRP adds average order value, product variety, return rate and spending trend.

## 4.2 Clustering algorithms

- **K-Means:** fast and interpretable, but assumes compact groups and is sensitive to scale and outliers.
- **DBSCAN:** finds dense regions and marks outliers as noise without a fixed k, but is sensitive to its parameters.

RFM variables are skewed, so CIRP applies a log transform and scaling, then compares K-Means with DBSCAN.

## 4.3 Choosing and validating clusters

Without labels, clusters are judged by the Silhouette coefficient, the Davies-Bouldin index and the elbow plot. CIRP also profiles each segment (mean RFM, size, churn rate) and gives it a business name and a suggested action.

# 5. Research Gap and Positioning of CIRP

Earlier work on e-commerce customers segmented them with K-Means and then predicted churn, reporting better results after segmentation. CIRP tests this by comparing the churn model with and without the segment as a feature. Most reviewed work uses contract-based telecom data; CIRP uses non-contractual retail data and combines both models in one reproducible application.

# 6. Lecturer Feedback & Evaluation

*Feedback & Evaluation and Final Grading Criteria are reserved for the lecturer.*

## Feedback & Evaluation

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## Suggested Improvements

- **Value-based targeting:** rank at-risk customers by churn probability times expected spend, so the retention budget goes first to customers who are both likely to leave and worth keeping.
- **Customer-specific churn window:** wholesale customers order irregularly, so a fixed 90-day label can mark a normal gap as churn; define churn relative to each customer's own typical time between orders.
- **Segment migration tracking:** use the 5 snapshots to see how customers move between segments and treat a move to a weaker segment as an early warning signal.
- **Reasons per prediction:** show the top factors behind each customer's churn risk in the app, so the business sees why and can choose the right action.
- **New customers:** customers with a single order behave differently from repeat buyers; handle them with a separate rule or model instead of the same one.

## Final Grading Criteria

| Criterion | Maximum marks | Marks awarded |
|---|---|---|
| Documentation | | |
| Implementation | | |
| Testing | | |
| Presentation | | |
| **Total** | | |